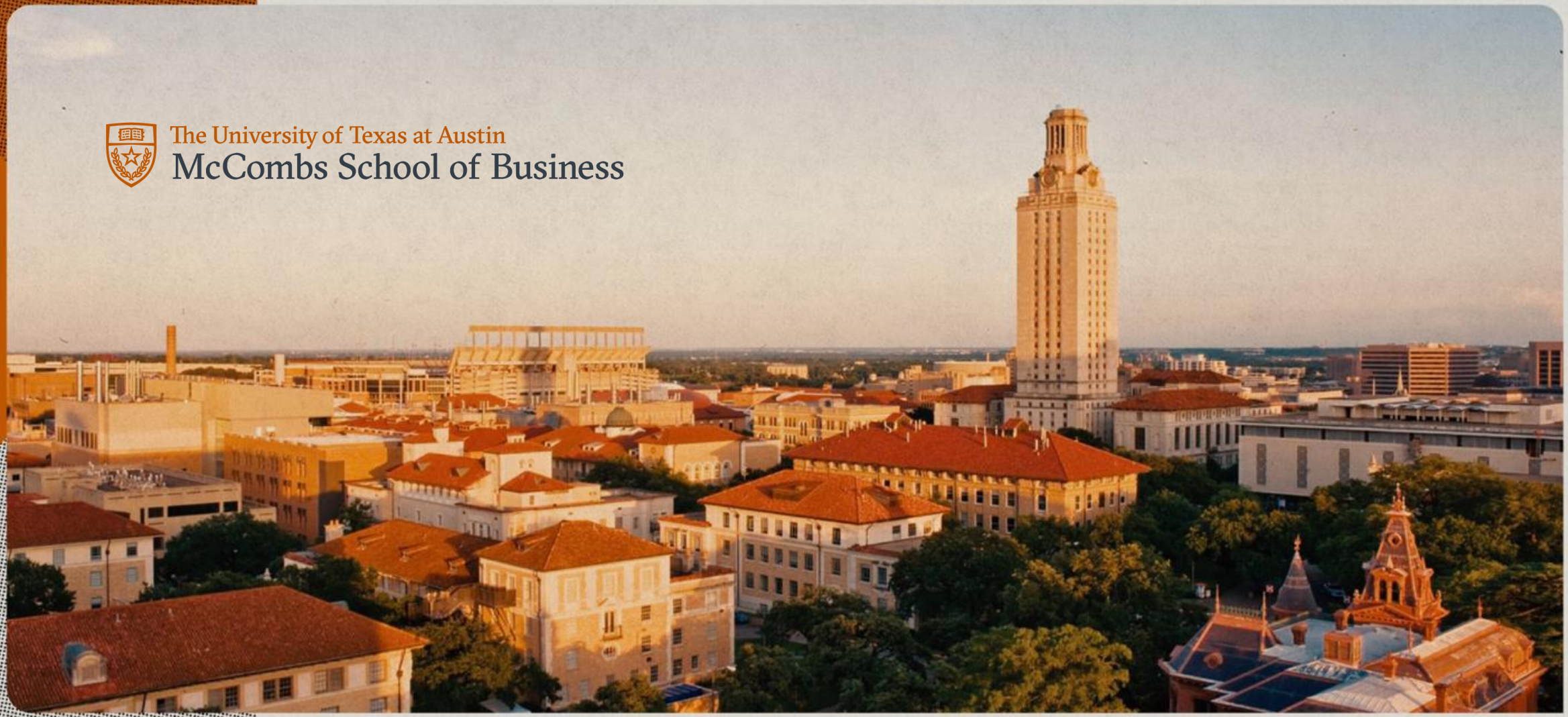


D5: Cluster model selection

Learning Objectives

1. Understand heuristics for selecting number of clusters:
2. Elbow method
3. Silhouette score
4. Gap statistic
5. Graphical overlap of PC1,PC2 plots

Major R commands

fviz_cluster => plots PC2 vs PC1 for a proposed *k*

fviz_nbclust => scree plot for Elbow method

clusGap => computes Gap statistic for range of *k*'s

fviz_gap_stat => plots Gap statistic for range of *k*'s

Agenda

1. Discuss heuristics for number of clusters
2. Heuristics demo using Austin homes sample

   random100 (including Zone and defs).xlsx
3. Your turn

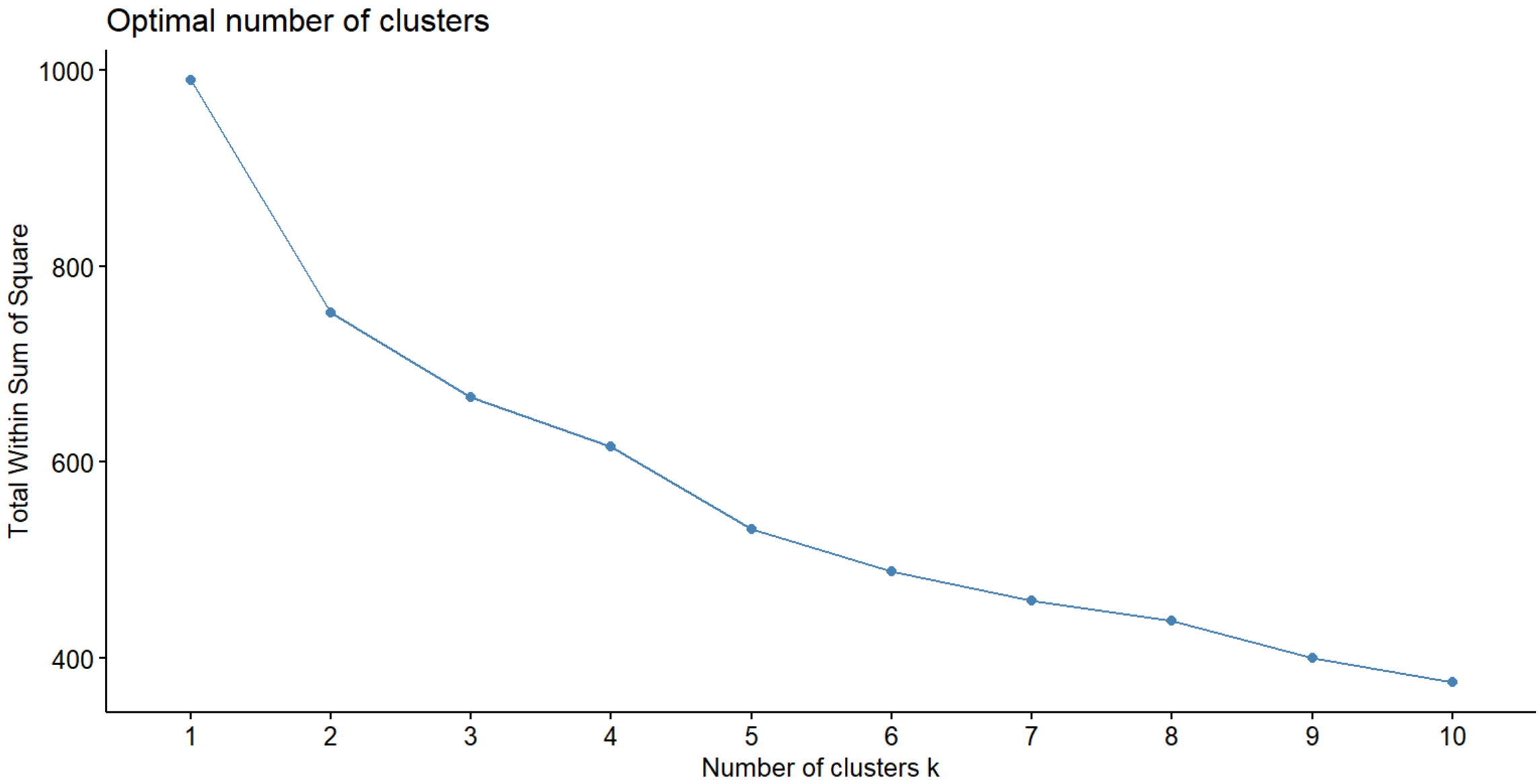
Intro

<from *Cluster Analysis Notes (choosing K).pdf* >

Elbow method

- Balances explanatory power vs complexity.
- WSS(k) is the variability of the clustering variables that is unexplained by the k clusters, and BSS(k) is the explained variability.
- Since WSS(k) + BSS(k) = TSS (constant for all k), then the plot of WSS(k) vs k shows how the unexplained part declines with increasing complexity.
- Where the plot levels off (the “elbow”), further gains in explanation are not worth the added complexity.

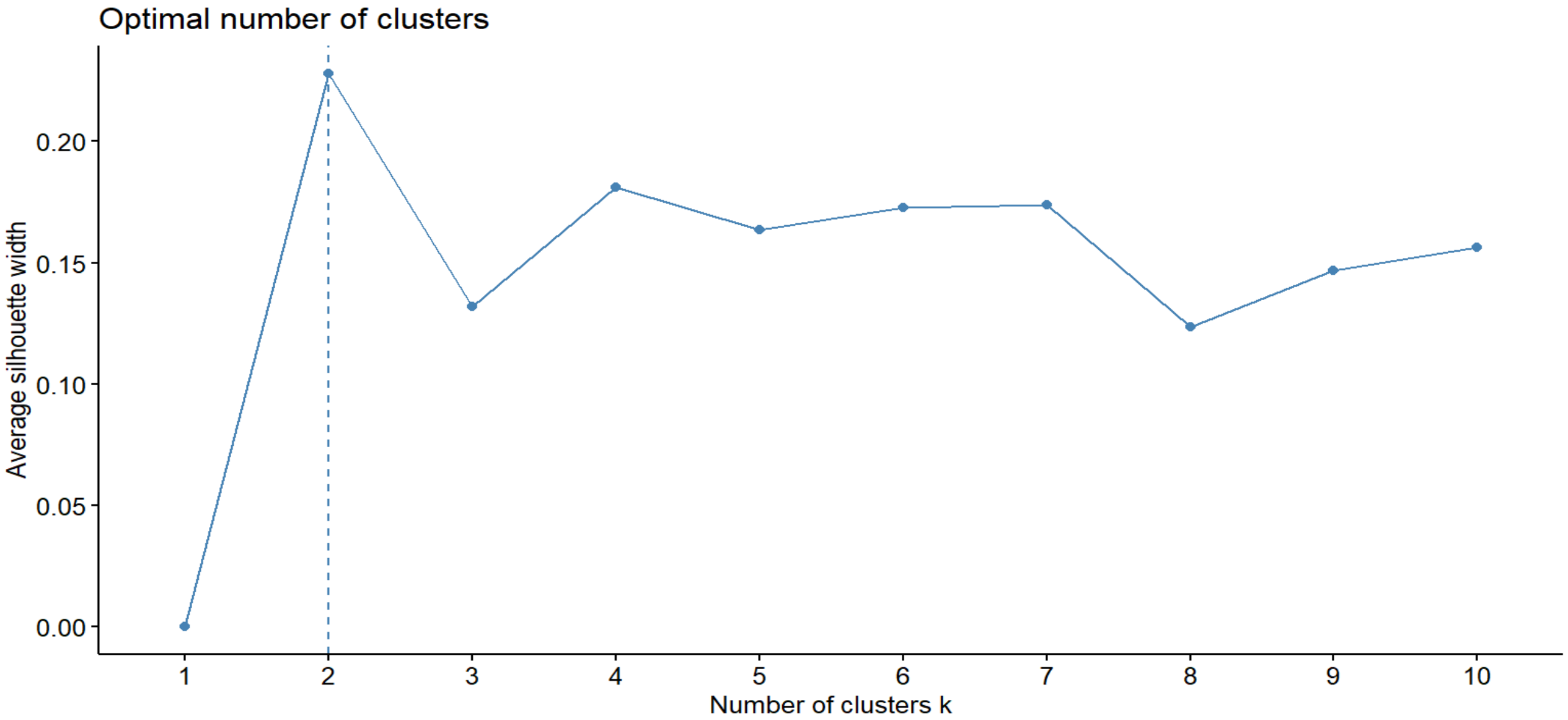
Elbow Method



Average Silhouette method

- Shows within-cluster similarity vs between-cluster dissimilarity.
- a(i) is mean distance of case i to other cases in i's cluster
- b(i) is mean distance from case i to the cases in closest cluster to i
- s(i) = b(i) – a(i), scaled by max of a(i), b(i) [**silhouette**]
- **Average silhouette** is mean of s(i) over all cases in the dataset
- A good clustering should have a large average silhouette.
- Average silhouette lies between -1 and +1.
- **Average silhouette method** plots average silhouette vs *k* & picks max

Silhouette Method



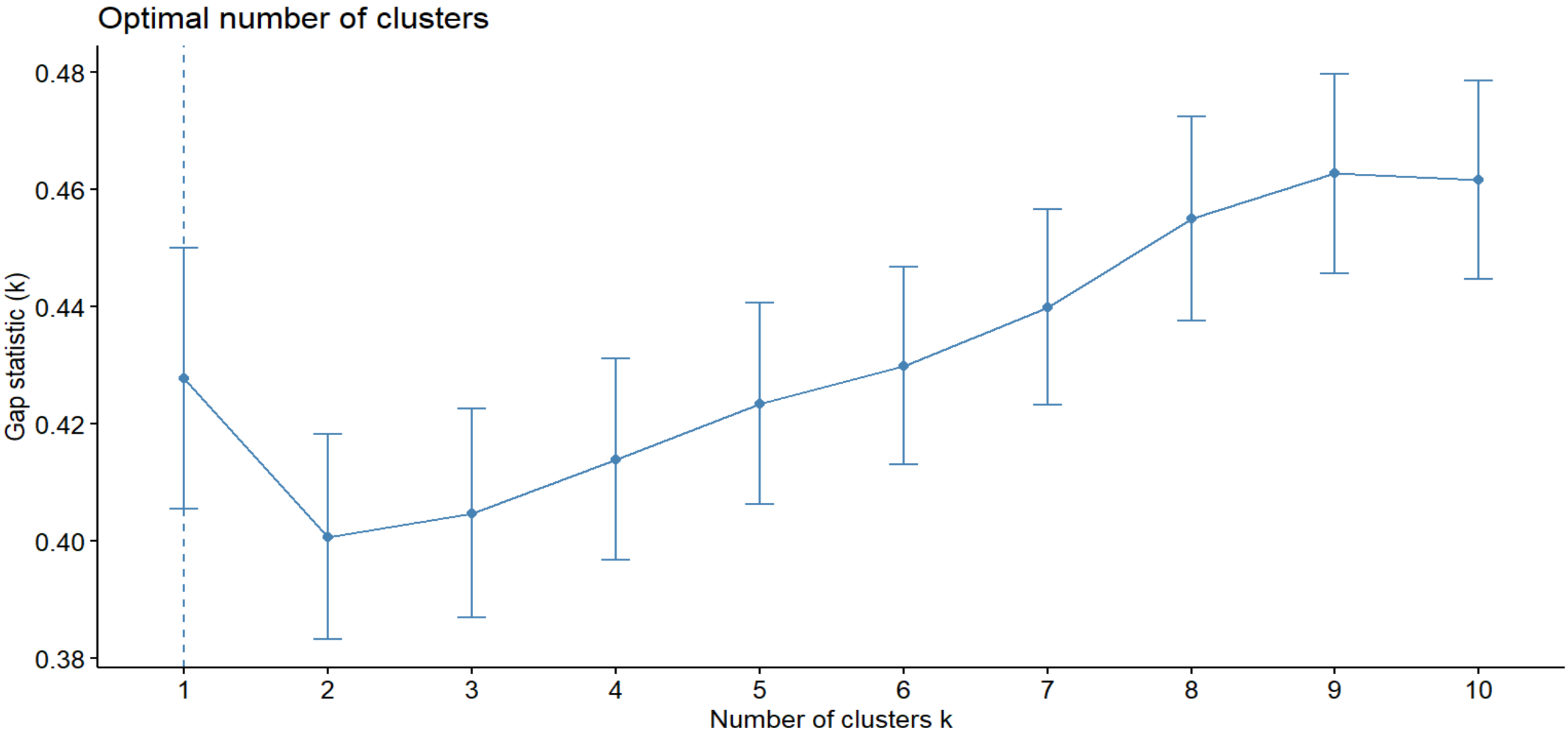
Gap Statistic method

- Gap statistic method compares actual WSS(k) to the value expected for WSS(k) if there are no clusters
- Rationale: If there are no real clusters, then any clusters that are found are just due to chance configurations of the data. Then any found clustering should not explain much, and BSS should be small, and WSS should be large.
- The Gap Statistic is the difference between expected and actual values of WSS(k). Choose k where the difference (gap) is greatest.

Gap Statistic (technical details)

- The Gap Statistic is calculated by simulating the WSS(k) of a flat, featureless distribution of data and averaging the results.
- For technical reasons, log(WSS(k)) is used instead of WSS(k). This makes the Gap statistic more stable.
- Since the Gap statistic tends to increase and then level off, the authors of the method suggest selecting a kind of elbow in the plot: Choose the smallest k for which the increase < one standard deviation s_{k+1} of the simulated log(WSS(k+1)) values: i.e., Gap(k+1) – Gap(k) > s_{k+1}.

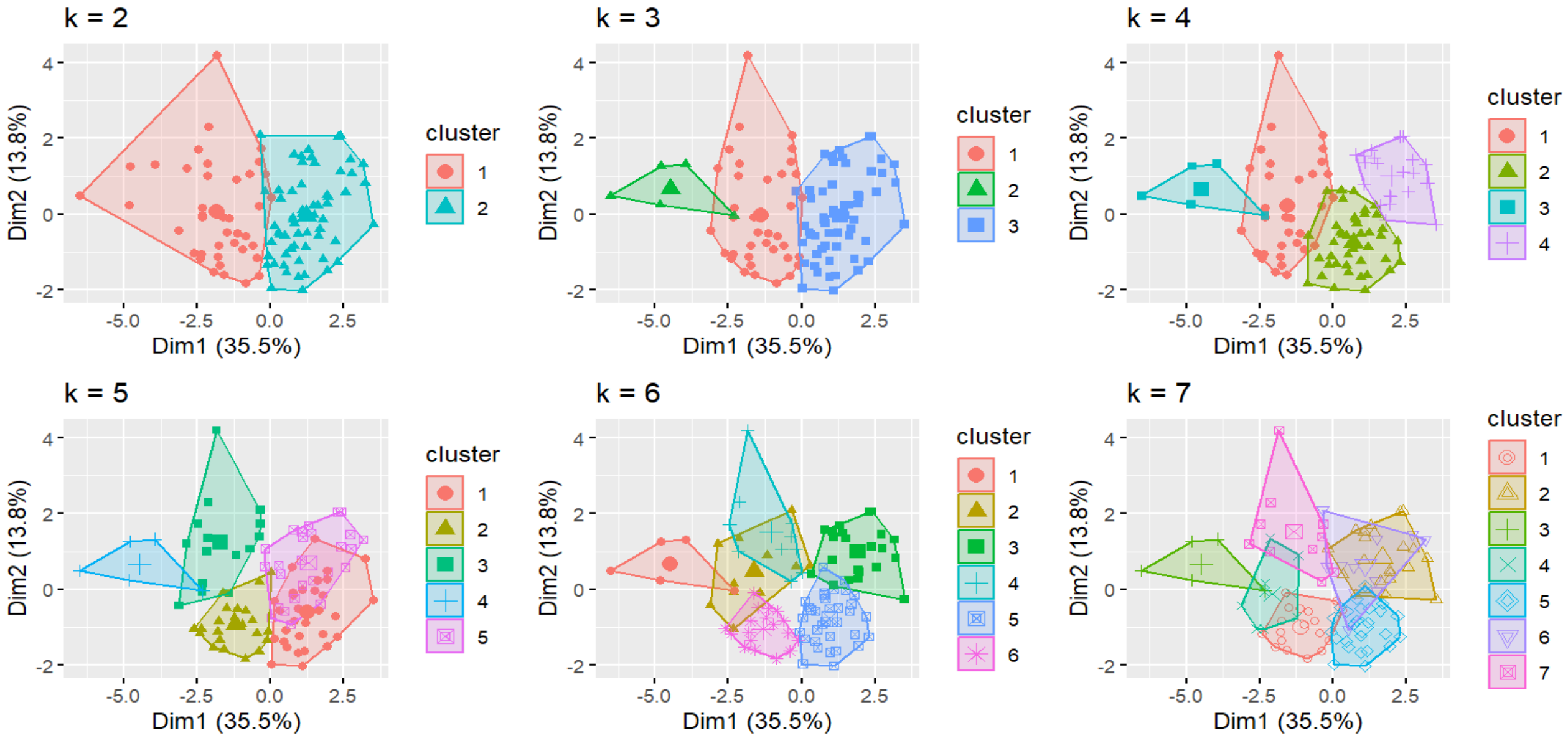
Gap Statistic Method



Visual Plots of the Principal Components of the Clustering Variables

- **fviz_cluster** converts the data values into principal components and plots PC2 vs PC1 for all data points, then organizes the plot into color-coordinated clusters for a specified value of k.
- This helps show how separated the clusters are in terms of the data dimensions that matter most.
- Visually distinct clusters suggest a good value for k.
- Visually overlapping clusters suggest a poor value for k.
- But apparently distinct (overlapping) clusters can be overlapping (distinct) on higher-order, less important PC's.

Cluster Overlap (PC2 vs PC1)



Your turn

Use all variables in *AustinApartmentRent.xls* including Rent.

Use standardized variables.

Use the Elbow, Average Silhouette, and Gap statistic methods to find recommendations for the number of clusters.